Stopping Criteria

Absolute Error:

$$\varepsilon^{(k)} = \|\mathbf{x}^* - \mathbf{x}\|_2 = \left[\sum_{i=1}^{n}(x_i^* - x_i)^2\right]^{1/2}$$

$$= \sqrt{(x_1^* - x_1)^2 + (x_2^* - x_2)^2 + \cdots + (x_n^* - x_n)^2}$$

$\mathbf{x}^* =$ Exact Solution

$\mathbf{x} =$ Numerical Solution

$$\varepsilon^{(k)} = \left\|\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)}\right\|_2$$

$\mathbf{x}^{(k)} =$ Numerical Solution from k^{th} Iteration

Relative Error:

$$\varepsilon^{(k)} = \frac{\|\mathbf{x}^* - \mathbf{x}\|_2}{\|\mathbf{x}^*\|_2} = \frac{\sqrt{(x_1^* - x_1)^2 + (x_2^* - x_2)^2 + \cdots + (x_n^* - x_n)^2}}{\sqrt{(x_1^*)^2 + (x_2^*)^2 + \cdots + (x_n^*)^2}}$$

$$\varepsilon^{(k)} = \frac{\left\|\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)}\right\|_2}{\|\mathbf{x}^{(k)}\|_2}$$

Stopping Criteria: $\varepsilon^{(k)} < \varepsilon_t$

$\varepsilon_t =$ Specified Tolerance

Relative Residual:

$$\varepsilon^{(k)} = \frac{\left\|\mathbf{r}^{(k)}\right\|_2}{\|\mathbf{r}^{(0)}\|_2}$$

$\mathbf{f}(\mathbf{x}) = 0$

$\mathbf{r}^{(0)} = \mathbf{f}(\mathbf{x}^{(0)})$

$\mathbf{r}^{(k)} = \mathbf{f}(\mathbf{x}^{(k)})$

ε_t of 10^{-6} and 10^{-8} will be used in this course.